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RESEARCH

Noninvasive CT radiomics-clinical model accurately classifies anhydrous uric acid stones: a multicenter study

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Abstract

Background Urolithiasis, particularly anhydrous uric acid stones (AUAs), imposes significant clinical and economic burdens. Accurate preoperative differentiation of AUAs from other stone types remains challenging, yet essential for personalized patient management.

Methods In this multicenter retrospective study, 468 patients diagnosed with urolithiasis underwent preoperative non-contrast CT imaging. Radiomics analysis was performed, extracting 1,218 features that were subsequently reduced to six key features via LASSO regression. A combined predictive model was developed by integrating radiomics-derived probabilities with independent clinical predictors. The model's performance was rigorously assessed through internal and external validation datasets using ROC analysis, calibration plots, decision curve analysis, and SHAP for model interpretability.

Results The combined radiomics-clinical model achieved excellent diagnostic performance with area under the ROC curve (AUC) values of 0.942 (training set), 0.944 (internal test set), and 0.957 (external validation set), significantly surpassing individual clinical or radiomics models. SHAP analysis identified urine pH, serum uric acid, and radiomics-derived predicted probability as critical predictive factors for AUAs. Calibration and decision curve analyses supported the model's robust predictive reliability and potential for meaningful clinical benefit.

Conclusion This study demonstrates that integrating CT-based radiomics with key clinical variables significantly improves preoperative differentiation of AUAs, providing a robust, non-invasive diagnostic tool that can enhance personalized treatment strategies. Future research should aim to incorporate metabolomic profiling and advanced imaging modalities to further optimize clinical translation and implementation.

Keywords Urolithiasis · Anhydrous uric acid stones · Radiomics · CT imaging · Machine learning · Personalized medicine · SHAP analysis

Introduction

Urolithiasis is one of the most prevalent benign urological disorders worldwide, with a rising incidence that poses substantial healthcare challenges, including significant economic costs and increased patient morbidity [1, 2]. Clinical management and therapeutic decision in urolithiasis are largely influenced by stone composition, which determines whether patients require invasive surgical intervention or can benefit from conservative pharmacological treatment [3, 4]. Uric acid stones (UAs) account for approximately 10–15% of urinary calculi but have historically received less research attention than calcium-based stones, despite their distinct therapeutic implications [5]. Anhydrous uric acid stones (AUAs), the predominant subtype of UAs, represent the most thermodynamically stable crystalline form of uric

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acid and exhibit notable physicochemical stability, making them resistant to conventional dissolution treatments [6–8].

Accurate preoperative differentiation between AUAs and non-anhydrous uric acid stones (nUAs) is essential for personalized treatment planning. While AUAs typically respond well to urinary alkalinization and hydration therapies—allowing patients to avoid invasive surgical procedures—nUAs often require combination treatments, including surgical intervention [9]. Current clinical guidelines recommend infrared spectroscopy as the definitive method for stone compositional analysis; however, its use is confined to postoperative specimens and is therefore unsuitable for preoperative planning [10]. Although dual-energy computed tomography (DECT) has emerged as a promising non-invasive imaging modality for preoperative stone characterization, its broader clinical application is limited by higher radiation doses, elevated costs, and restricted availability in resource-constrained settings [11].

Radiomics, an advanced computational technique for extracting and quantifying high-dimensional imaging features, has significantly transformed diagnostic and prognostic approaches in oncology by identifying imaging biomarkers predictive of disease behavior and treatment response [12–14]. Radiomics provided novel tools to support complex clinical decision-making [12–14]. Kim et al. identified stone density and urinary pH as important predictors for distinguishing UA stones, while Spettel et al. used Hounsfield unit measurements and urine pH to achieve high predictive accuracy for UA stones [15, 16]. Nonetheless, the application of radiomics in differentiating AUA from nUA remains unexplored, representing a critical gap in current clinical practice [17–19].

This study therefore aimed to develop and rigorously validate an integrated radiomics-clinical predictive model using preoperative CT imaging and clinically relevant variables. We hypothesized that this combined model could accurately distinguish AUAs from nUAs, thereby supporting precision management strategies in urolithiasis. By integrating multidimensional radiomics features with conventional clinical variables, this research seeks to enhance non-invasive diagnostic accuracy, contribute to personalized patient care, and address a methodological gap with potential implications for improving clinical outcomes.

Methods

Patients data collection

This retrospective multicenter study was approved by the institutional review boards of Huai'an First People's Hospital of Nanjing Medical University (No. KY-2024-070-01) and the Affiliated Taizhou People's Hospital of Nanjing Medical University (No. KY-2021-030-01). Due to the retrospective nature, the requirement for informed consent was waived. A total of 560 patients diagnosed with urolithiasis who underwent non-contrast CT between April 2021 and December 2024 at the two institutions were initially reviewed. The inclusion criteria were: (1) confirmed diagnosis of urinary stones, (2) surgical stone removal performed, (3) postoperative stone composition confirmed as AUAs or nUAs via infrared spectroscopy, and (4) availability of high-quality preoperative thin-slice CT images (slice thickness = 1 mm). Exclusion criteria included: (1) concurrent urological conditions (e.g., tumors, congenital malformations), (2) presence of urinary drainage devices (e.g., nephrostomy tubes, double-J stents, urinary catheters), and (3) poor-quality CT images with significant artifacts or incomplete clinical data. After applying these criteria, 468 patients were included in the final analysis. Patients from Huai'an First People's Hospital (n = 418) were randomly allocated into a training cohort (n = 292, 70%) and an internal test cohort (n = 126, 30%). Patients from the Affiliated Taizhou People's Hospital (n = 50) comprised the external validation set. Clinical and demographic data were extracted from electronic medical records using a prespecified data dictionary. Age was calculated on the date of CT acquisition and sex was recorded. Height and weight values recorded nearest to the CT (within 7 days) were used to calculate body mass index (BMI, kg/m^2). Stone location was categorized on imaging as renal pelvis/calyx, proximal/mid/distal ureter, or bladder, and the maximal stone diameter was measured on axial images or multiplanar reformats (millimeters). Preoperative blood tests performed within 7 days of CT were recorded, including serum creatinine, estimated glomerular filtration rate (eGFR, calculated by the CKD-EPI equation), blood urea nitrogen, serum calcium, phosphate, uric acid, and C-reactive protein; all assays were performed in the participating hospitals' clinical laboratories using standard automated methods and reported according to local reference ranges. Urinalysis data included urine pH, specific gravity, dipstick results, microscopic red and white blood cell counts (microscopic hematuria defined as >3 RBC/HPF), and urine culture when available (positive culture defined as >10^5 CFU/mL). If multiple measurements fell within the specified window, the value closest to the CT date was used. Data abstraction and entry were performed independently by two investigators, and any discrepancies were adjudicated by a third investigator. For missing data, a complete-case approach was used when the missing proportion was low; otherwise, multiple imputation by chained equations was applied in sensitivity analyses. The proportion of missing values for each clinical variable is summarized in

Supplementary Table 1, whereas radiomics features derived from CT images had no missing entries.

CT imaging protocol

Non-contrast CT examinations were performed on multi-slice CT scanners at both institutions using a standardized protocol. All images were reconstructed with a routine soft-tissue convolution kernel. Additional scanner-specific details are provided in Supplementary Table 2.

Stone composition analysis

Stone composition was analyzed in the participating hospitals' clinical laboratories using an automated infrared spectroscopic analyzer (LIIR-20, Lamoride, Tianjin, China) according to the manufacturer's instructions. Stone fragments were handled as part of routine clinical workflow: specimens were rinsed, air-dried and pulverized before measurement, and spectra were matched against the instrument reference library to yield percent composition of each chemical component. Routine reporting was performed by medical laboratory technologists experienced in stone analysis, and results were reviewed by senior laboratory personnel or a clinically qualified reviewer as per local practice. For descriptive purposes, a single probability threshold was applied across datasets when calculating accuracy, sensitivity, and specificity; however, the current study did not establish a definitive clinical cut-off for treatment decisions, such as initiating urinary alkalinization or surgical planning. For the purposes of this retrospective study, the spectrometer reports (percent composition and peak assignments) were re-examined centrally by a study investigator; when available the investigator cross-checked the laboratory report against the raw spectral output. A dominant component was defined as any single constituent that accounted for >50% of the stone composition. Stones were grouped for analysis as AUAs or nUAs according to the study classification. Quality control was ensured by use of the manufacturer's calibration procedures and the laboratory's routine QA/QC processes.

Image segmentation and preprocessing

Image segmentation was performed using ITK-SNAP software (Version 4.0; http://www.itksnap.org). A radiologist with 10 years of experience manually delineated the stone boundaries to define the regions of interest (ROIs), carefully excluding adjacent renal parenchyma, vessels, perirenal fat, and imaging artifacts. All ROIs were independently reviewed by a senior radiologist with 20 years of experience to ensure consistency and accuracy. Any discrepancies were resolved through consensus discussion. To assess segmentation reproducibility, 30 CT scans from the training set were randomly selected and independently segmented twice by both radiologists at a two-week interval. Intraclass correlation coefficients (ICCs) >0.80 were considered indicative of high intra- and inter-observer agreement. Prior to radiomics feature extraction, all segmented images were resampled to isotropic voxel dimensions of 1 mm × 1 mm × 1 mm using linear interpolation to reduce variability caused by differences in CT scanner resolution.

Radiomic feature extraction and selection

A total of 1,218 radiomics features were extracted from each ROI using the PyRadiomics software (version 3.0.1; https://pyradiomics.readthedocs.io/en/latest/), adhering to the Image Biomarker Standardization Initiative (IBSI). These features included first-order statistics, shape-based features, and texture features derived from gray-level dependence matrices (GLDM), gray-level co-occurrence matrices (GLCM), gray-level run-length matrices (GLRLM), and gray-level size zone matrices (GLSZM). To capture multiscale texture information, wavelet-transformed features were also generated (Supplementary Appendix S2). All features were standardized using Z-score normalization. Feature selection was conducted in four steps: (1) initial statistical screening using Student's t-tests or Mann-Whitney U tests ($P < 0.05$), (2) removal of redundant features based on Pearson correlation analysis ($r > 0.80$), (3) ranking of remaining features via the minimum redundancy maximum relevance (mRMR) algorithm, and (4) final selection of key predictive features using Least Absolute Shrinkage and Selection Operator (LASSO) regression with 10-fold cross-validation. This multistep selection process aimed to reduce overfitting and improved model generalizability.

Model construction and selection

Based on the selected radiomics features, five classifiers were developed and compared: LASSO-penalized logistic regression, support vector machine (SVM), random forest (RF), decision tree (DT), and naïve Bayes (NB). Radiomics features were standardized (z-score) prior to model development. For clinical model development, potential predictors were screened by univariable logistic regression ($P< 0.10$) and then entered into a multivariable LASSO-penalized logistic regression using the glmnet package (alpha = 1). Optimal penalty parameter (λ) was selected by 10-fold cross-validation in the training cohort and the more parsimonious λ.1se was used to define the final feature set with nonzero coefficients. The radiomics model output (predicted probability) was combined with the selected clinical predictors to construct an integrated model; the integration step was performed using multivariable logistic regression on the training set to yield a combined prediction score and to facilitate coefficient interpretation and nomogram construction. Model discrimination was assessed by receiver operating characteristic (ROC) curves and area under the curve (AUC), with 95% confidence intervals estimated by bootstrap resampling (n = 1000). Sensitivity, specificity, and accuracy were reported at the prespecified or optimal Youden threshold. Calibration was evaluated with calibration plots and the Hosmer–Lemeshow test, and clinical usefulness was assessed by decision curve analysis. Model performance was reported separately in the internal test set and the external validation set. In order to address class imbalance, we employed stratified train–test splitting and stratified 10-fold cross-validation to preserve the proportion of AUAs and nUAs across all folds. No additional class-weighting or resampling techniques were applied. To further evaluate model performance under class imbalance, precision-recall (PR) analysis was conducted on the external validation set.

Model evaluation

Model discrimination performance was comprehensively assessed and compared using ROC curves. Differences in AUC between models were statistically evaluated via the DeLong test. Calibration of the integrated model was accessed using the Hosmer-Lemeshow goodness-of-fit test and calibration curves. Decision curve analysis (DCA) was employed to evaluate the net clinical benefit and potential utility of the model in real-world decision-making contexts.

Model interpretability

To improve transparency and interpretability of the predictive model, Shapley Additive exPlanations (SHAP) analysis was conducted. Global feature importance was visualized using SHAP summary plots, while individualized feature contributions were illustrated with SHAP force plots, providing intuitive explanations of feature influence on model predictions.

Statistical analysis

Statistical analyses were performed with SPSS (version 25.0, IBM Corp.) and R (version 4.4.1, https://www.r-project.org). Continuous variables were assessed for normality using the Shapiro–Wilk test and are presented as mean ± standard deviation (SD) for normally distributed data or median with interquartile range (IQR) for non-normally distributed data; categorical variables are presented as counts and percentages. Group comparisons used Student's t test or the Mann–Whitney U test for continuous variables and the chi-square test or Fisher's exact test for categorical variables, as appropriate. For multivariable clinical model development we applied LASSO-penalized logistic regression using the glmnet package (alpha = 1). Predictor variables were standardized prior to modeling. Optimal penalty parameter (λ) was chosen by 10-fold cross-validation in the training cohort and the more parsimonious λ.1se was used to select features with nonzero coefficients; the final penalized model coefficients are reported. Model discrimination was assessed by area under the receiver operating characteristic curve (AUC) using the pROC package, with 95% confidence intervals obtained by 1000 bootstrap resamples. Calibration was examined by calibration plots and the Hosmer–Lemeshow goodness-of-fit test using the rms and ResourceSelection packages. Clinical utility was evaluated by decision curve analysis using the dca.R script. Model performance was reported separately in the internal test set and the external validation set. In addition to ROC analysis, precision–recall (PR) curves and PR areas under the curve (PR-AUCs) were generated for each model in the training, internal test, and external validation sets. Missing data were handled by complete-case analysis when the proportion of missingness was negligible; otherwise multiple imputation by chained equations (mice package) was performed and included in sensitivity analyses. All tests were two-sided and a P value < 0.05 was considered statistically significant.

Results

Clinical characteristics

Figure 1 depicts the multicenter patient flow and cohort split. It shows the initial screened cohort, the exclusion steps, and the final 468 included cases. It also shows the random split at Huai'an into training (n = 292) and internal test (n = 126), and the external validation cohort from Taizhou (n = 50). A total of 468 patients (336 males, 71.8%; 132 females, 28.2%) were enrolled, with 137 (29.3%) diagnosed as AUAs and 331 (70.7%) as nUAs. Detailed clinical demographics and laboratory characteristics are summarized in Table 1. Patients in the AUAs group were significantly older (57.42 ± 10.85 vs. 50.09 ± 12.62 years, p< 0.001), had higher BMI (26.39 ± 3.70 vs. 25.55 ± 3.59, p = 0.002), and exhibited notably lower urinary pH values (5.84 ± 0.44 vs. 6.24 ± 0.45, p< 0.001) compared to those in the nUAs group. Additionally, significant differences were identified in serum uric acid (406.86 ± 117.71 vs. 333.01 ± 89.90 µmol/L, p< 0.001), serum creatinine (133.28 ± 100.64 vs. 82.84 ± 55.41 µmol/L, p< 0.001), triglycerides (2.01 ±

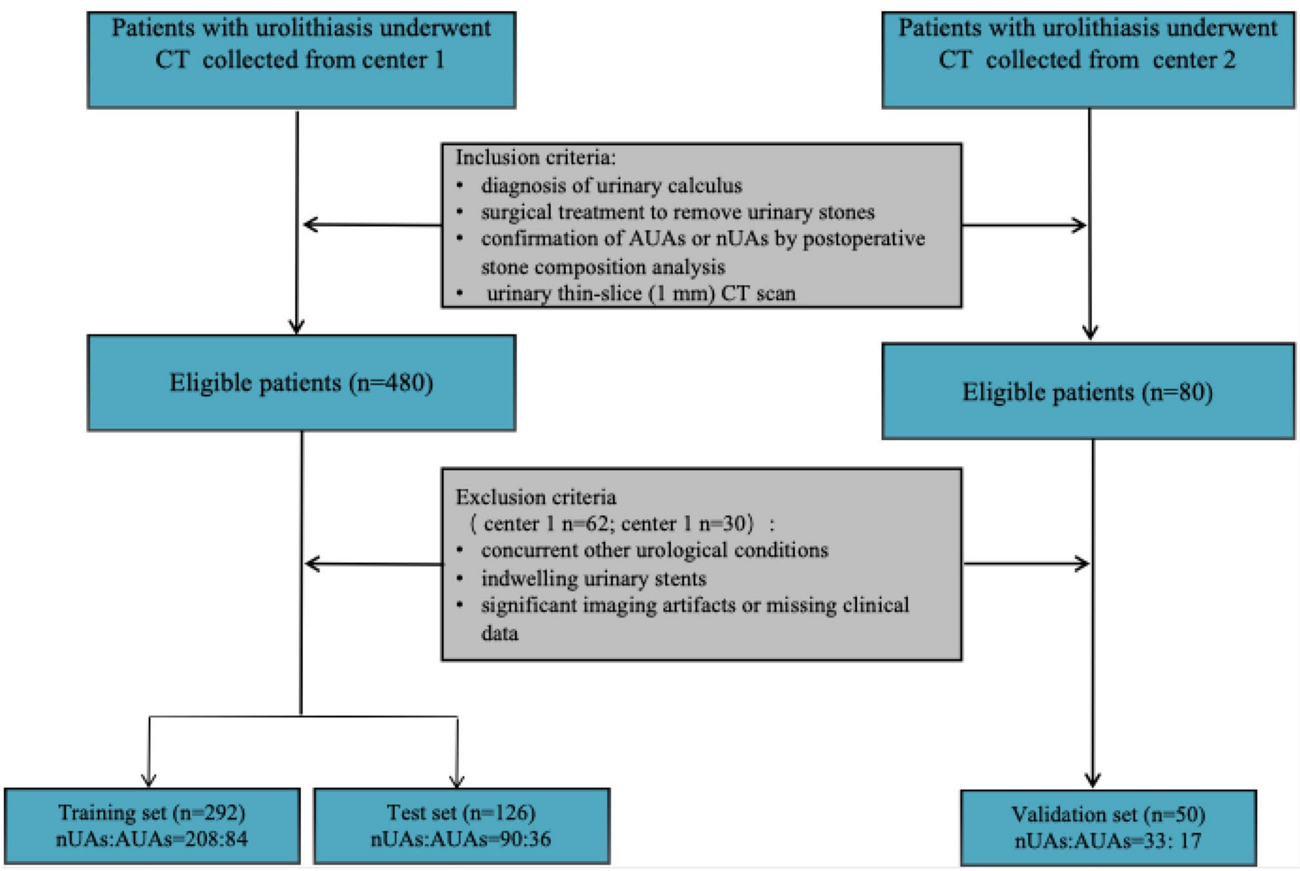

Fig. 1 Multicenter patient flow, eligibility, and train–test–validation split

1.63 vs. 1.75 ± 1.13 mmol/L, $p = 0.007$), fasting blood glucose (5.85 ± 2.00 vs. 5.38 ± 1.60 mmol/L, $p < 0.001$), and urine white blood cells (WBC; 137.98 ± 459.97 vs. 474.29 ± 1991.96 cells/µL, $p = 0.003$).

Figure 2 summarizes the end-to-end modeling workflow. It shows stone ROI segmentation on non-contrast CT, extraction of 1,218 radiomics features, stepwise reduction to six features, and training of radiomics, clinical, and combined models. It also shows the evaluation by ROC, calibration, and decision-curve analysis, and the model explanation with SHAP.

Radiomics feature selection and model development

Of the initial 1218 extracted radiomics features, 1208 redundant or irrelevant features were excluded through Pearson correlation (>0.80) and the mRMR algorithm. The remaining 10 features were entered into LASSO regression, from which six features exhibiting non-zero coefficients were ultimately selected (Supplementary Fig. S1a–b). Details of the selected radiomics features and their corresponding coefficients are provided in Supplementary Table S3. ROC curves analysis of the five machine learning classifiers (LASSO, RF, SVM, DT, NB) demonstrated that the LASSO algorithm achieved the highest diagnostic performance, with AUC values of 0.921 (95% CI: 0.884–0.957) in the training set and 0.923 (95% CI: 0.859–0.987) in the internal test set (Fig. 3; Table 2).

Performance comparison among radiomics, clinical, and combined models

Model-predicted probabilities differed significantly between AUAs and nUAs groups (violin plots in **Supplementary Fig. S2**; $p < 0.001$). Univariate and multivariate logistic regression analyses identified urine pH, age, BMI, serum uric acid, serum creatinine, triglycerides, and urine culture results as independent clinical predictors. These variables were integrated with the probability outputs of the radiomics model to construct a combined radiomics–clinical model.

Across all datasets, the combined model exhibited superior discriminatory performance compared to both the radiomics-only and clinical-only models. In the internal test set, the combined model achieved an AUC of 0.944 (95% CI: 0.892–0.996), significantly superior to the clinical model (AUC=0.827, 95% CI: 0.859–0.987; $p < 0.001$). The radiomics model showed a trend towards superior

Table 1 Clinical baseline data of patients with urolithiasis

Characteristic	AUAs (n = 137)	nUAs (n = 331)	P-value
Sex, n (%)			0.503
Female	30 (21.9%)	102 (30.8%)	
Male	107 (78.1%)	229 (69.2%)	
Age, mean ± SD	57.42 ± 10.85	50.09 ± 12.62	< 0.001
Location, n (%)			0.171
UB	9 (6.6%)	8 (2.4%)	
LK	41 (29.9%)	94 (28.4%)	
LU	31 (22.6%)	71 (21.5%)	
RK	42 (30.7%)	76 (23.0%)	
RU	13 (9.5%)	81 (24.5%)	
Urethra	1 (0.7%)	1 (0.3%)	
Hypertension, n(%)			0.067
N	85 (62.0%)	245 (74.0%)	
Y	52 (38.0%)	86 (26.0%)	
BMI, mean ± SD	26.39 ± 3.70	25.55 ± 3.59	0.002
WBCs, mean ± SD	6.85 ± 1.91	6.63 ± 1.86	0.379
FBG, mean ± SD	5.85 ± 2.00	5.38 ± 1.60	< 0.001
SUA, mean ± SD	406.86 ± 117.71	333.01 ± 89.90	< 0.001
TG, median ± SD	2.01 ± 1.63	1.75 ± 1.13	0.007
SCr, median ± SD	133.28 ± 100.64	82.84 ± 55.41	< 0.001
Urine PH, median ± SD	5.84 ± 0.44	6.24 ± 0.45	< 0.001
Urine WBC, median ± SD	137.98 ± 459.97	474.29 ± 1991.96	0.003
Urine RBC, median ± SD	1168.29 ± 7836.94	963.79 ± 5154.59	0.875
UCx			0.023
Negative	87(63.5%)	252(76.1%)	
Positive	50(36.5%)	79(23.9%)	
Urinary urease			0.125
Negative	66(48.2%)	119(36.0%)	
Positive	71(51.8%)	212(64.0%)	

UB Urinary Bladder, *LK* Left kidney, *LU* Left ureter, *RK* Right kidney, *RU* Right ureter, *BMI* Body Mass Index, *WBCs* White Blood Cells, *FBG* Fasting Blood Glucose, *SUA* Serum uric acid, *TG* Triglycerides, *SCr* Serum Creatinin, *UCx* Urine Culture

performance compared to the clinical model (AUC difference = 0.096, p = 0.054, Table 3a), though this did not reach conventional statistical significance. No statistically significant difference was observed between the combined and radiomics models (AUC difference = −0.021, p = 0.466, Table 3a). In the training set, the combined model achieved an AUC of 0.942 (95% CI: 0.914–0.971), significantly outperforming the clinical model (AUC=0.857, 95% CI: 0.810–0.903; p< 0.001) and marginally surpassing the radiomics model (AUC=0.921; p = 0.016). A statistically significant, albeit small, performance difference was observed between the combined and radiomics models (AUC difference = −0.021, p = 0.031, Table 3b), favoring the combined model. In the external validation set, the combined model achieved an AUC of 0.957 (95% CI: 0.905–1.000), significantly outperforming the clinical model (AUC=0.750, 95% CI: 0.606–0.895; p = 0.001) and the radiomics model (AUC=0.954, 95% CI: 0.902–1.000; p = 0.761). Both the combined and radiomics models demonstrated significantly superior discriminatory performance compared to the clinical model alone (p < 0.01), but no statistically significant difference was observed between the combined and radiomics models (AUC difference = −0.004, p = 0.761, Table 3c). Fig. 4 shows the ROC curves for the combined, radiomics, and clinical models in the training set, internal test set, and external validation set. The curves highlight the superior performance of the combined model compared to the clinical model, with marginal differences when compared to the radiomics model in the validation sets.

To further assess model performance under class imbalance, precision-recall (PR) curves were generated for the combined, radiomics, and clinical models on the external validation set. The PR curves, shown in Supplementary Figure 3, depict the performance of the combined model (Com_validation, green), radiology model (Rad_validation, orange), and clinical model (Cil_validation, purple). The shaded areas around the curves represent the 95% confidence intervals derived from bootstrapping. These results reveal a favorable precision-recall trade-off for all three models, with the combined model demonstrating the highest average precision (AP). Supplementary Table 4 summarizes the corresponding PR performance metrics, including the AP values and their 95% confidence intervals. The PR curves and AP values align with the ROC-AUC results, further supporting the robustness and clinical utility of the combined model in accurately distinguishing AUAs from nUAs under conditions of class imbalance.

The external validation cohort comprised N AUAs and M nUAs, and 95% confidence intervals for AUC, sensitivity, specificity, and accuracy are reported in Table 4.

Calibration and clinical utility assessments further supported the superiority of the combined model. The calibration plot demonstrated excellent agreement between predicted probabilities and observed outcomes (Hosmer-Lemeshow test, p>0.05; Fig. 5a). Decision curve analysis revealed the combined model consistently provided the highest net clinical benefit across most probability thresholds compared with either individual models (Fig. 5b).

Combined model interpretation using SHAP analysis

Model interpretability was evaluated using SHAP analysis. The SHAP global bar plot identified the predicted probability from radiomics features, urine pH, age, serum uric acid, BMI, and serum creatinine as the most influential predictors (Fig. 6a). The SHAP summary beeswarm plot illustrated the

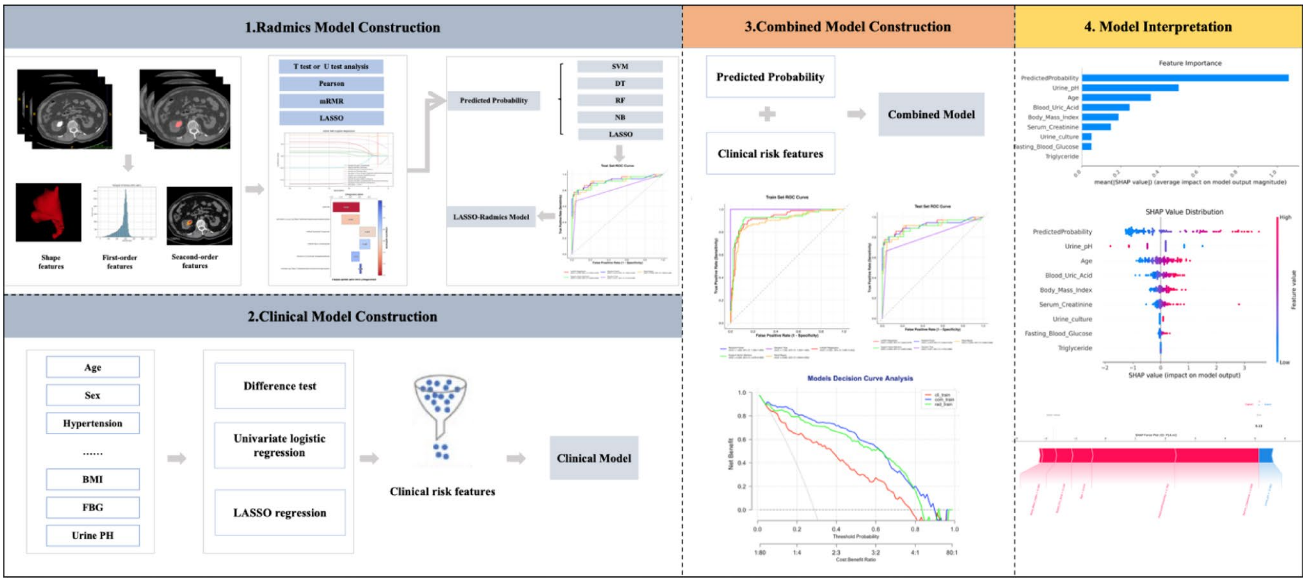

Fig. 2 Workflow for building the radiomics, clinical, and combined models with SHAP interpretation

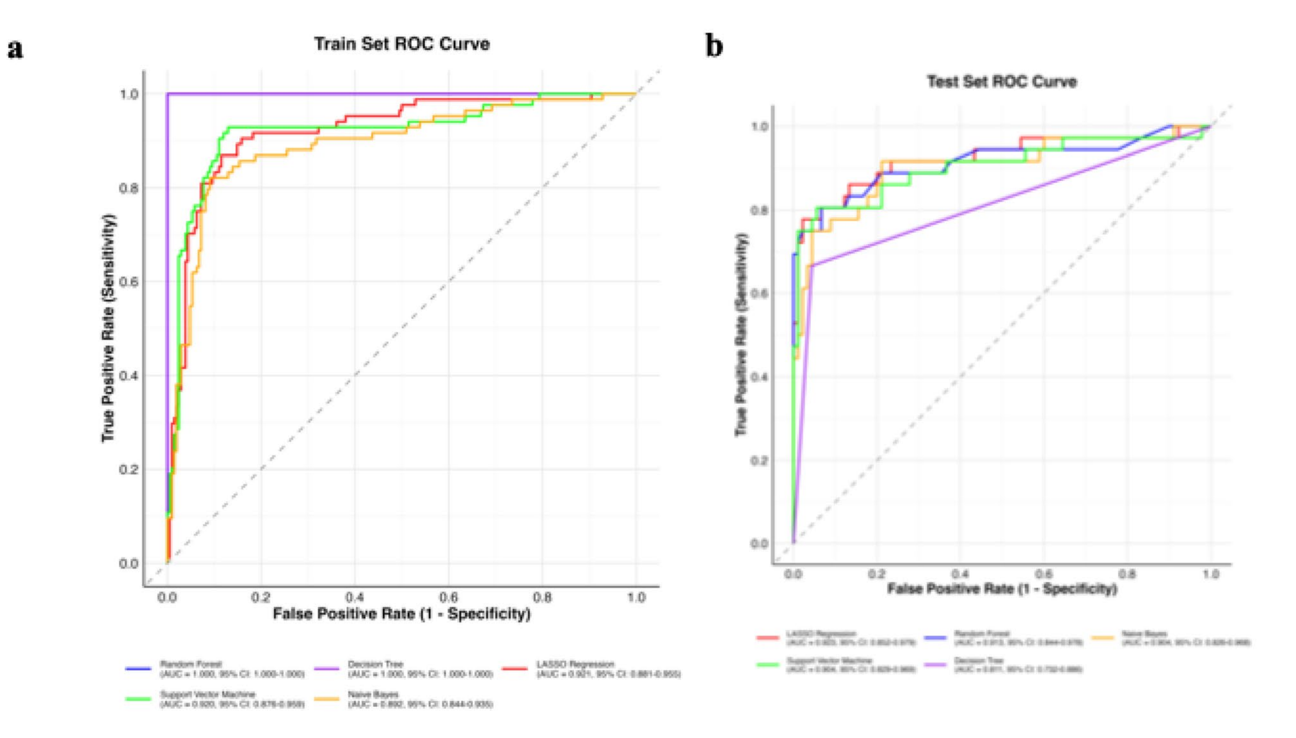

Fig. 3 ROC curves of five machine learning algorithms in the training set (**a**), test set (**b**). ROC receiver operating characteristic

Table 2 Diagnostic performance of five machine learning algorithms in training and test sets

	Training set					Test set				
Model	AUC	95%CI_	Sensitivity	Specificity	Accuracy	AUC	95%CI_	Sensitivity	Specificity	Accuracy
LASSO	0.921	0.881–0.955.881.955	0.869	0.885	0.88	0.923	0.852–0.979.852.979	0.778	0.978	0.921
RF	1.000	1.000-1.000.000.000	1.000	1.000	1.000	0.913	0.844–0.978.844.978	0.750	0.978	0.913
SVM	0.920	0.876–0.959.876.959	0.929	0.870	0.887	0.904	0.829–0.969.829.969	0.806	0.944	0.905
DT	1.000	1.000-1.000.000.000	1.000	1.000	1.000	0.811	0.732–0.886.732.886	0.667	0.944	0.865
NB	0.892	0.844–0.935.844.935	0.821	0.909	0.884	0.904	0.826–0.968.826.968	0.750	0.956	0.897

LASSO Least absolute shrinkage and selection operator, *SVM* support vector machine, *DT* decision tree, *RF* random forest, *NB* Naive Bayes

Table 3 Delong's test for pairwise model comparison on the (a) internal test set, (b) training set, (c) external validation set

Comparison	Model 1	Model 2	AUC (Model 1)	AUC (Model 2)	AUC Difference	*p*
(a)						
Clinical vs. Combined	Clinical Model	Combined Model	0.827	0.944	0.117	< 0.001
Clinical vs. Radiomics	Clinical Model	Radiomics Model	0.827	0.9234	0.096	**0.054**
Combined vs. Radiomics	Combined Model	Radiomics Model	0.944	0.923	−0.021	0.466
(b)						
Clinical vs. Combined	Clinical Model	Combined Model	0.857	0.942	0.086	< 0.001
Clinical vs. Radiomics	Clinical Model	Radiomics Model	0.857	0.921	0.064	0.016
Combined vs. Radiomics	Combined Model	Radiomics Model	0.942	0.921	−0.021	0.031
(c)						
Clinical vs. Combined	Clinical Model	Combined Model	0.750	0.957	0.207	**0.001**
Clinical vs. Radiomics	Clinical Model	Radiomics Model	0.750	0.954	0.203	**0.003**
Combined vs. Radiomics	Combined Model	Radiomics Model	0.957	0.954	−0.004	0.761

AUC area under the curve, *CI* confidence interval, *PPV* positive predictive value, *NPV* negative predictive value, *F1* F1-score

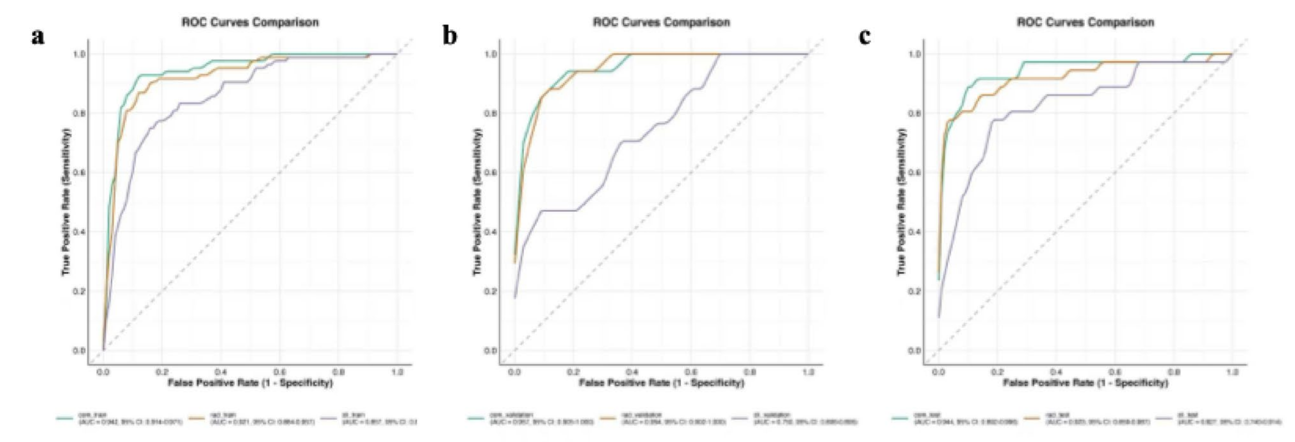

Fig. 4 ROC curves of the radiomic model, clinical model, and combined model in predicting AUAs in the training set (**a**), test set (**b**), and validation set (**c**). ROC receiver operating characteristic,*AUC* area under the curve, AUAs anhydrous uric acid stones

Table 4 Detailed performance metrics on the external validation set (*n* = 50, 17 positive cases)

Model	Best Threshold	Sensitivity (95% CI)	Specificity (95% CI)	Youden Index
Cli_validation	0.823	0.471 (0.176 - 0.706)	0.939 (0.576 - 1.000)	0.410
Com_validation	0.693	0.882 (0.588 - 1.000)	0.909 (0.606 - 1.000)	0.791
Rad_validation	0.328	0.882 (0.529 - 1.000)	0.909 (0.686 - 1.000)	0.791

impact of feature values on predictions, revealing a clear pattern: higher predicted probability from the radiomics model and elevated serum uric acid levels strongly favored classification as AUAs, whereas higher urine pH values were strongly associated with nUAs classification (Fig. 6b). Patient-specific SHAP force plots provided individualized explanations by visualizing how each feature contributed to a given prediction. In representative cases, elevated predicted probabilities and serum uric acid levels drove AUAs classification, while increased urine pH strongly counteracted this effect (Fig. 7).

Overall, these results underscore the robust diagnostic potential and clinical utility of integrating CT-based radiomics features with clinical variables for preoperative differentiation of AUAs, offering valuable insights into individualized patient management.

Discussion

The present study developed and rigorously validated a comprehensive radiomics-clinical model for the preoperative differentiation of AUAs from nUAs using CT imaging data combined with clinical parameters. The combined model demonstrated excellent diagnostic performance, achieving AUC values of 0.942, 0.944, and 0.957 in the training, internal test, and external validation sets, respectively. These

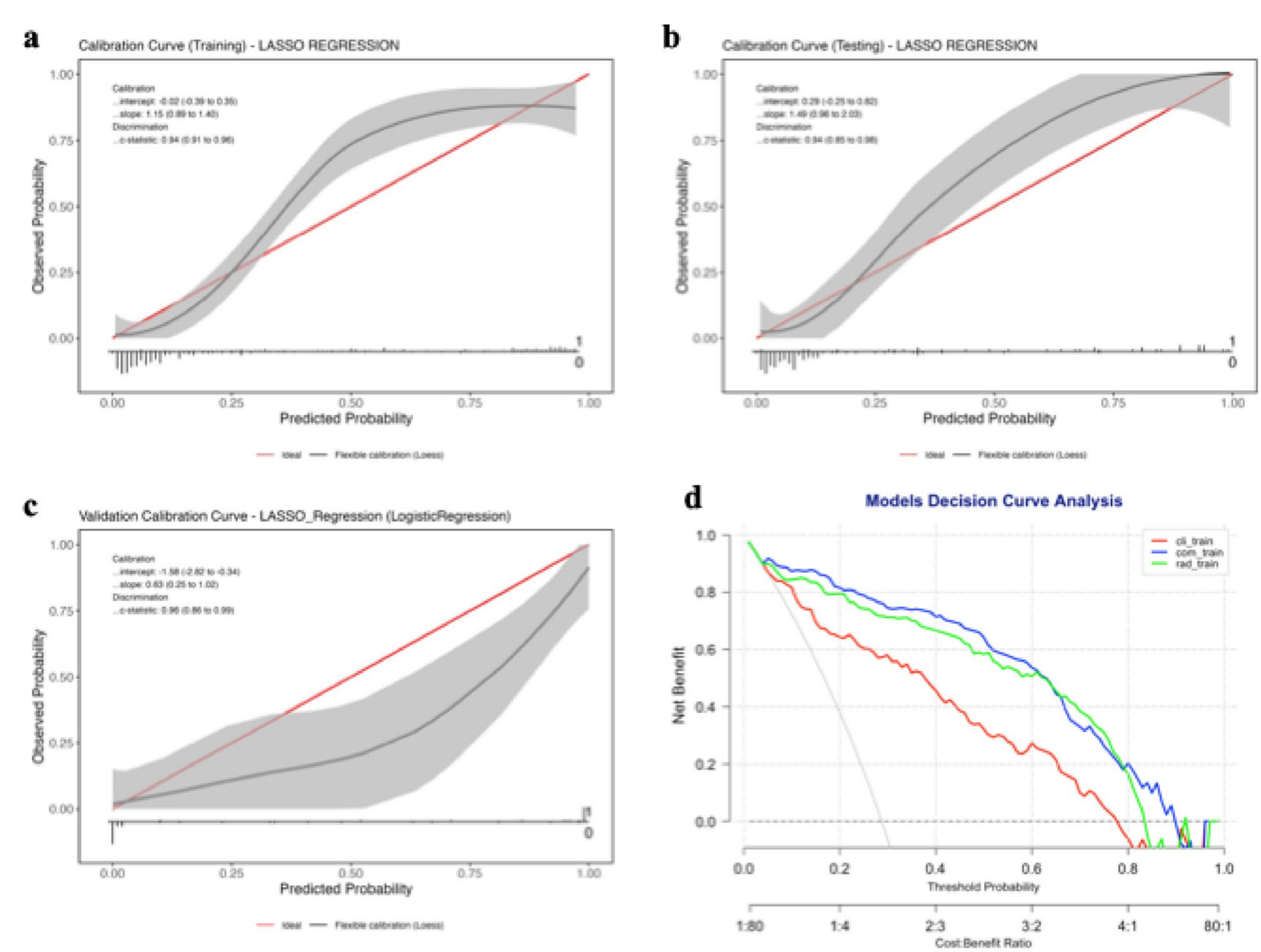

Fig. 5 Development and performance of combined model. **a** Calibration curve between the predicted and actual incidences of AUAs. **b** Decision curve analysis compares the net benefits of four scenarios in predicting the risk of AUAs: Combined model (blue line), Clinical model (red line), radmics model(green line), and All (gray line, refers to the assumption that all patients have AUAs). AUAs anhydrous uric acid stones

results significantly outperformed those of models based solely on radiomics or clinical models, underscoring the strength and effectiveness of integrating multidimensional imaging features with relevant clinical variables.

The superior performance of the combined model can be attributed to several underlying scientific mechanisms. Radiomics analysis captures microstructural differences in stone composition that is often overlooked by conventional imaging interpretation. By extracting high-dimensional quantitative features—such as texture and morphological characteristics—from CT images, radiomics provides detailed insights into the physicochemical variability characteristic of different stone subtypes. This heterogeneity correlates with thermodynamic stability and solubility profiles, particularly in AUAs, which are primarily crystalline and demonstrate resistance to conventional dissolution therapies. These findings align with prior theoretical frameworks suggesting that radiomics features effectively quantify microscopic differences reflective of distinct crystallization pathways [20].

Furthermore, incorporating critical clinical variables—particularly urine pH, serum uric acid, BMI, and serum creatinine—markedly enhanced the discriminative capacity of the model. Among these, urinary pH emerged as a dominant independent predictor. Reduced urinary pH promotes urate crystallization by lowering its solubility, thereby facilitating stone aggregation and growth. This biochemical mechanism is well-supported by previous epidemiological and experimental study [18]. Similarly, elevated serum uric acid, indicative of hyperuricosuria, aligns with established metabolic pathways implicated in uric acid stone formation, further strengthening the model's physiological plausibility.

Compared with previous radiomics-based studies, the present research offers notable theoretical and practical advancements. Prior studies employing conventional single-energy CT achieved only moderate classification

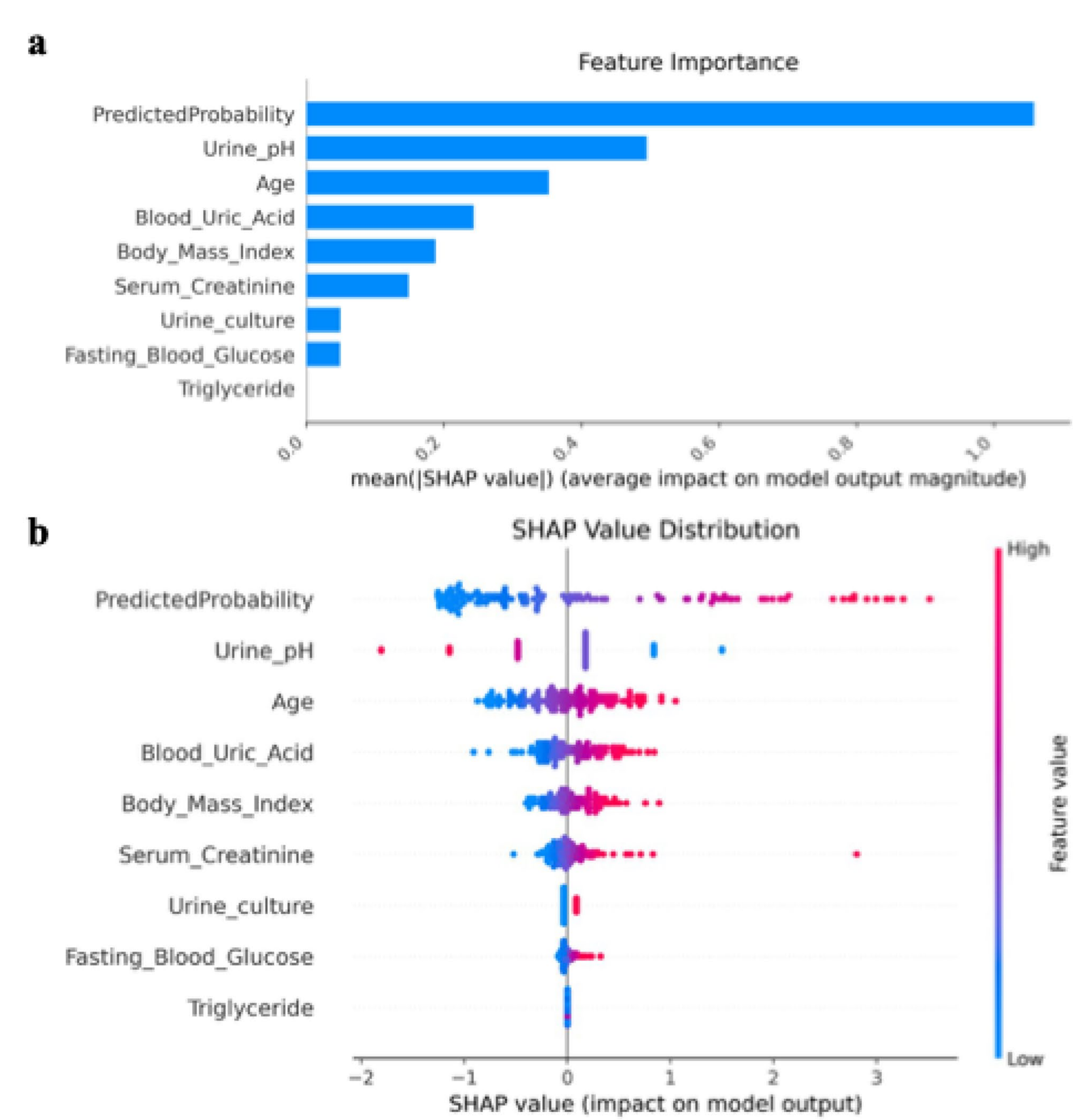

Fig. 6 Global visualization of the combined model through SHAP. (a) The SHAP bar plot shows the weights of the most important features in the model. (b) The SHAP beeswarm plot displays an information-dense summary illustrating the impact of top features on the model's output

performance for uric acid stone (AUC ~0.780), lacking specificity in differentiating AUAs from generalized uric acid stones [21]. In contrast, our integrated approach substantially improves both diagnostic accuracy and subtype specificity (AUC up to 0.957) [22]. Additionally, our model outperforms the infectious stone radiomics model proposed by Zheng et al. (AUC = 0.898) [23], highlighting the considerable benefit of incorporating comprehensive clinical parameters alongside advanced imaging analytics.

Despite these innovations, several methodological and interpretative limitations should be acknowledged. First, the limited number of AUAs and the relatively small external validation cohort may restrict generalizability, and prospective validation in larger populations is warranted. Second, the radiomics model, despite demonstrating excellent

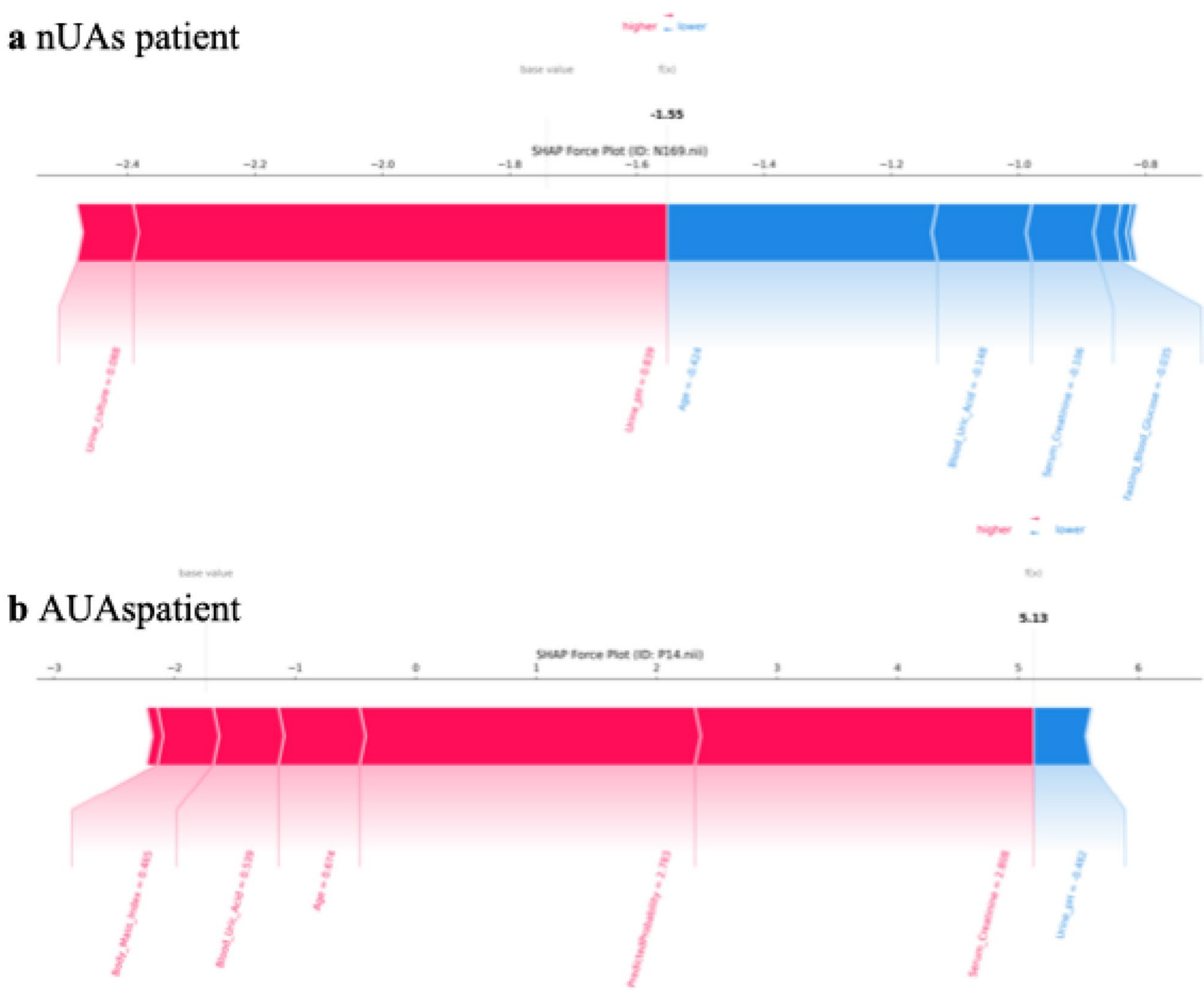

Fig. 7 SHapley Additive exPlanation (SHAP) force plot for two selected patients

diagnostic capabilities, currently offers limited biological interpretability. Recent metabolomics studies indicate that metabolites such as succinate possess inhibitory effects on stone formation through anti-inflammatory pathways [24]. Our findings align with prior studies, such as those by Kim et al. and Spettel et al., which demonstrated that clinical parameters and HU measurements can predict UA stones. However, our radiomics-based approach, which captures more detailed image features, offers enhanced discrimination between AUAs and non-AUAs, potentially improving diagnostic accuracy over traditional methods [15, 16]. Furthermore, in prospective clinical use, threshold selection should be tailored to the relative importance of sensitivity versus specificity (e.g., favoring higher sensitivity if the goal is to identify most patients eligible for urinary alkalinization), and is best determined through multidisciplinary consensus and further prospective evaluation. Establishing a definitive treatment-triggering threshold will require validation in a larger, prospective cohort.

Future research directions should pursue a multidimensional predictive approach, incorporating advanced functional imaging modalities such as diffusion-weighted MRI or dual-energy CT to further elucidate stone composition at a molecular level. Additionally, exploring the integration of radiomics with metabolomics and proteomics could yield a more comprehensive biomarker panel for improved diagnostic precision and therapeutic decision-making. Automation of image analysis software and its integration into clinical workflows could significantly enhance the practical translation of these findings, enabling rapid and accurate preoperative stone characterization across varied healthcare settings.

In summary, this study offers significant advancements in both scientific insight and clinical application for differentiating AUAs through the novel integration of radiomics and clinical predictors. The proposed model represents a robust, non-invasive diagnostic tool, and continued refinement through multidisciplinary collaboration and technological innovation holds substantial potential to improve personalized management and clinical outcomes in patients with urolithiasis.

Conclusion

A combined radiomics-clinical model was developed to enable accurate preoperative differentiation between anhydrous uric acid stones (AUAs) and nUAs. By integrating imaging-derived features with key clinical variables, the model facilitates optimized treatment planning—supporting a useful non-invasive aid to preoperative decision-making, but its clinical application should be confirmed in future multicenter prospective studies. This model offers a reliable and clinically applicable tool to advance personalized care in the management of urolithiasis.

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Author contributions Zhaofei Shi, Zhen Xu, and Liming Zhang were responsible for study design, data collection, and initial manuscript drafting. Wei Zheng contributed to clinical data acquisition and image analysis. Xuzhong Liu supervised the clinical interpretation, provided critical revisions, and ensured methodological rigor. Lili Guo conceived and supervised the overall study, contributed to study design, manuscript drafting, and final approval of the version to be published. All authors reviewed and approved the final manuscript.

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Data availability The data that support the findings of this study are available on request from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval Ethical approval was obtained from the Ethics Committee of the Affiliated Huaian No. 1 People's Hospital of Nanjing Medical University, KY 2021-030-01. This retrospective study was approved by the institutional review boards of both participating centers, and the requirement for individual informed consent was waived because only de-identified data were analyzed.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish Consent for publication was obtained from all participants.

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